

The National Residue Survey (NRS) is delivered by the department and monitors residues in animal products through various random and targeted testing programs.

Random residue monitoring includes 19 meat programs, an egg program, a honey program, and two seafood programs.

The random programs are designed to:

- ensure participating industries satisfy Australian export certification and importing country requirements
- enable domestic meat processing facilities to satisfy state and territory government regulatory authority licensing requirements
- provide evidence of good practice in the use of pesticides and veterinary medicines by the participating industries
- support quality assurance initiatives in participating industries.

Targeted animal product residue monitoring programs are designed to meet particular management objectives or monitor potential chemical residues that could pose a risk for access to export or domestic markets.

All animal product residue monitoring programs are designed, operated and reviewed, within agreed budgets, by the NRS in consultation with peak industry bodies.

Random monitoring programs

Of the meat programs, cattle, sheep, goats and pigs provide the largest number of samples for residue monitoring. Other meat programs cover products derived from camels, buffalo, eggs, deer, goats, honey, horses, kangaroos, poultry (chicken, duck, turkey, spatchcock and quail), and wild boar. The two seafood programs cover aquaculture and wild-caught seafood.

Sample collection and analysis

Collection rates are based on national production levels for each commodity, or, in the case of exported products, on market access requirements.

Sample collections at production or processing establishments is overseen by official sample collectors. The number of samples taken per establishment is proportional to their processing volume. The department allocates the required samples to each participating establishment and specifies the production period during which samples are to be taken. Animals are then selected for sampling at random. Once samples are collected, they are sent to a central receipt and dispatch facility within the department for a quality assurance check before they are forwarded to appropriate laboratories for analysis.

Choosing material for analysis

The matrix selected for analysis is the one that is expected to contain the highest concentration of a residue. The matrix may be inedible and does not necessarily represent the part most likely to be eaten. For example, fat is analysed for pesticides, kidney is analysed for antibiotics, liver is analysed for metals, and urine or faeces is analysed for some hormonal growth promotants.

Chemical screens

The choice of chemicals for measurement in the samples is guided by the likelihood of residues from pesticides, veterinary medicines or contaminants. The chemicals include those used commonly in agricultural and veterinary practice, as well as those necessary to fulfil export requirements. Some chemicals monitored are not registered for use in Australian animal production systems, nor are likely to be used, but may be important to satisfy the requirements of international trading partners.

The range of chemical screens applied to animal product samples is listed in the following table:

Veterinary drugs and animal treatments

Group	Chemicals
Anthelmintics	Benzimidazoles, closantel, macrocyclic lactones and triclabendazole
Antibiotics	Aminoglycosides, anticoccidials, antimicrobials, beta lactams, cephalosporins, macrolides, nitroimidazoles, phenicols, sulphonamides, fluoroquinolones, quinolones and tetracyclines
Hormones	Resorcylic acid lactones, steroids, stilbenes and trenbolone
Other veterinary drugs	Beta-agonists, corticosteroids, sedatives, andro and non-steroidal anti-inflammatory drugs

Pesticides, animal treatments and environmental contaminants

Group	Chemicals
Fungicides, herbicides and insecticides	Benzoyl ureas, carbamates, fungicides, herbicides, insecticides, organochlorines, organophosphates, persistent organic pollutants and pyrethroids
Environmental Contaminants	Metals

Random monitoring program results for 2023-24

Random monitoring program results for 2023-24

Commodity	Samples Compliance rates (%)	
Camel	15	100

Cattle	5,337	99.98
Buffalo	11	100
Chicken	308	100
Deer	7	100
Game deer	10	100
Hen eggs	258	99.22
Goat	271	99.63
Honey	157	99.36
Horse	29	100
Kangaroo	50	100
Pigs	1,011	99.70
Poultry (other than meat chicken and hen eggs)	42	100
Sheep	2,113	99.81
Wild boar	71	100
Aquaculture seafood	187	99.47
Wild-caught seafood	107	100
TOTAL	9,984	99.87

**Cattle
program
summary
2013–14
to 2023–
24**

Year	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24
Samples	6,000	4,375	4,386	4,576	4,576	4,877	5,352	5,649	5,258	5,300	5,300
Compliance (%)	99.96	99.96	100.00	99.85	99.89	99.94	99.91	99.96	99.96	99.98	99.98

**Sheep
program
summary
2013–14
to 2023–
24**

Year	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021–22	2022-23	2023-24
Samples	5,494	2,543	2,539	2,590	2,591	2,589	2,682	2,905	2,009	2,309	2,113
Compliance (%)	99.76	99.68	99.68	99.20	99.69	99.73	99.78	99.86	99.45	99.91	99.81

**Pig
program
summary
2013–14
to 2023–
24**

Year	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021– 22	2022-23	2023– 24
Samples	2,763	2,759	2,751	1,020	1,002	1,000	1,020	1,000	1,014	1,007	1,011
Compliance (%)	99.92	99.51	99.40	99.20	99.41	99.40	98.80	99.24	99.51	99.40	99.70

Targeted monitoring programs

Samples included in target monitoring programs are collected and tested under the National Residue Survey in accordance with agreed industry requirements. Results are released to relevant authorities and to industry for action where necessary.

National organochlorine residue management (NORM) program

The NORM program focuses on minimising the risks of organochlorine (OC) residues in beef and is jointly funded by the beef industry and state/territory governments. Besides testing cattle from at-risk properties at abattoirs, the NORM program results assist owners of properties with identified OC contamination hazards to develop and apply on-farm property management plans to minimise the risk of OC residues.

The department coordinates the program and manages the payments to others involved, such as laboratories and state/territory governments.

National antibacterial residue minimisation (NARM) program

The NARM program focuses on minimising the occurrence of antibacterial residues in bobby calves from dairy farms and is funded by the beef industry. State/territory governments support the program through activities related to traceback investigation, and the management of dairy farms found to have consigned bobby calves for slaughter with antibacterial residues above relevant Australian standards.

Investigations have found that residue contraventions occur when management systems are inadequate or break down. Consequently, a major focus of activities is to work with industry quality assurance schemes and stakeholders to introduce initiatives that heighten farmer awareness and minimise the risk of residues occurring. The department coordinates the program and manages the payments to others involved, such as laboratories and state/territory governments.

Targeted antibacterial residue testing programs

There are several programs, each targeted to a different animal species. These programs focus on animals that present at establishments and are suspected by veterinary inspectors of having received antibacterial treatment inside the required withholding period. Each program combines targeted testing, quality assurance, extension and regulation to minimise antibacterial residues in beef.

The department coordinates each program and manages the payments to others involved, such as laboratories.

The programs are:

- Cattle targeted antibacterial residue testing (TART) programs
- Sheep targeted antibacterial residue testing (START) program
- Pig targeted antibacterial residue testing (PTART) program
- Horse targeted antibacterial residue testing (HTART) program
- Goat targeted antibacterial residue testing (GTART) program

Audit-related programs

Hormonal growth promotant (HGP) audit program

Some markets, such as the European Union (EU) and non-EU HGP sensitive markets, prohibit the importation of products from animals treated with HGPs. Australia has developed HGP-free programs that provide assurance to the market and allow Australian cattle producers to supply those markets.

On-farm audits are used to monitor compliance with accreditation requirements. The department manages the testing of samples taken during the audits and manages the payments to others involved such as laboratories, state/territory governments and other regulators.

Residue management audits

The NRS residue management audit program includes the cattle, sheep and goat industries. Since 2009, over 30,000 targeted property audits have been undertaken throughout Australia as part of a comprehensive approach to residue management.

Source: <https://www.agriculture.gov.au/agriculture-land/farm-food-drought/food/nrs/animal-residue-monitoring>